OPC UA

Industrial Control Systems · SCADA Hub

COURSE TOPICS

1. **OPC UA Architecture:** Client/server and pub/sub transport models, the transition from DCOM, and the platform-neutral unified namespace.
2. **Information Model:** Address space, node classes, NodeIDs, type definitions, and the self-describing browsing model.
3. **Services:** Read, Write, Browse, Subscribe, and CreateMonitoredItems — the full service set and which matter in practice.
4. **Sessions and Channels:** SecureChannel negotiation, session establishment, token lifetime, and reconnection behavior on channel loss.
5. **Security:** Security Mode None vs. Sign vs. SignAndEncrypt; certificate exchange; PKI structure; consequences of disabled security.
6. **Subscriptions:** Publishing interval, MonitoredItem sampling, deadband filters, and the configuration chain controlling network traffic volume.
7. **Data Types:** Scalars, structures, enumerations, arrays, and the StatusCode quality system for representing sensor validity.
8. **Discovery:** Local and Global Discovery Servers, endpoint enumeration, certificate trust, and open-discovery security implications.
9. **Implementation:** SDK selection, Ignition as OPC UA server, common integration failures, and Wireshark-based diagnostic workflow.
10. **Protocol Lab:** Full session capture — TCP connection, SecureChannel, browse, subscription, disconnect, and session token renewal.

LEARNING OUTCOMES

- Browse an OPC UA server's address space and interpret node classes and references.
- Configure Security Mode Sign and SignAndEncrypt with certificate exchange on a production server.
- Build and tune a subscription with appropriate publishing intervals and deadband filters.
- Verify security mode and session token behavior from a Wireshark packet capture.
- Select and configure an OPC UA SDK or integration platform for a given deployment scenario.

Certification Relevance

- **OPC Foundation Certified** — Developer and Integrator exams test the information model, service set, and security configuration directly.
- **GICSP (GIAC)** — OPC UA security modes and integration patterns appear in ICS security exam prep domains.
- **ISA CCST** — Industrial networking and system integration standards tested at all CCST levels.

Next Steps

1. Pair OPC UA with the Security guide for the strongest overlap with GICSP exam domains.
2. Download open62541 (free, open-source C stack) and run a local server against UA Expert client.
3. OPC Foundation spec Parts 1 (Concepts) and 4 (Services) are the normative reference.
4. Apply at opcfoundation.org for the OPC UA Integrator certification exam.